Temperature, Vapor Pressure Deficit, and Global Crop Yields: Evidence from a Multi-Crop Panel

Global evidence from eight staple crops: wheat, rye, barley, oats, maize, soybean, rice, sorghum. 1961–2024.

MAIN MESSAGE : Higher flowering-period temperature reduces yields across all eight crops, and **temperature effects extend beyond VPD.**

A. Conceptual Framework

Higher flowering-period temperature affects yields through both atmospheric dryness and direct heat stress

Higher Temperature
(flowering stage)

PATHWAY A
Higher VPD
Atmospheric dryness
Partial mechanism

PATHWAY B
Direct heat stress
Physiological damage
Additional mechanism

VPD-mediated yield loss

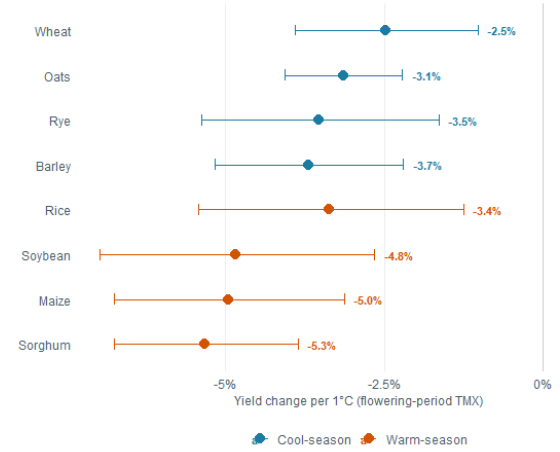
Total temperature effect (TMX)

−1.4% to **−3.0%** per 1°C

−2.5% to **−5.5%** per 1°C

B. Heat Sensitivity Across Crops

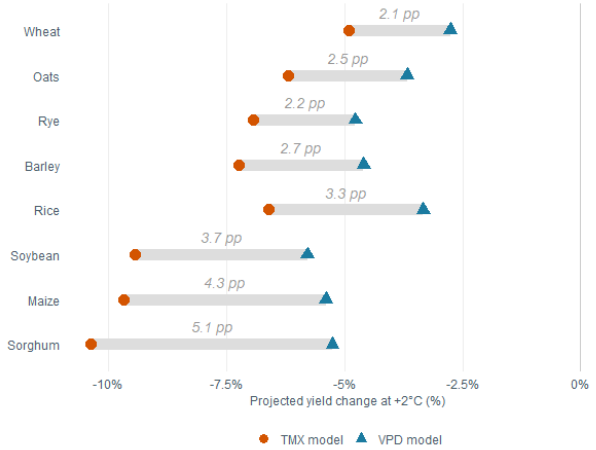
Estimated % yield change per 1°C increase in flowering-period TMX



Warm-season crops show the largest heat sensitivity

C. Temperature Effects Extend Beyond VPD

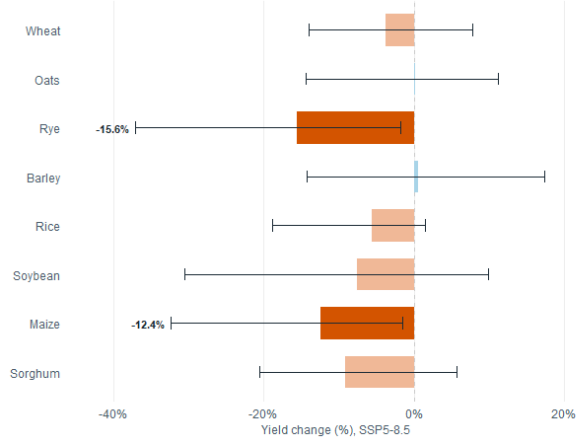
At +2°C warming, VPD explains only part of the temperature effect



TMX-based losses consistently exceed VPD-based losses

D. CMIP6 End-of-Century Projections

SSP5-8.5, 2081–2100, global production-weighted estimates (%)



Largest projected losses: rye, maize, and sorghum
Future warming implies substantial risks for major staple crops

CONCLUSION : Direct heat stress plays a substantial role beyond atmospheric dryness.

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